Main (Project/ Written) Assignment

Patient CareGuard: Temperature Monitoring and Fall Detection System

Embedded Processors UESTC & UESTCHN 2004 School of Engineering, University of Glasgow

UESTC & UESTC HN 2004 Main (Written) Assignment Assessment Criteria

Each individual Assignment will be evaluated using the below criteria:

- Demonstrating a functional embedded system project – 30%

  This will be assessed based on the video demonstration and your design implementation (software and hardware).

- Technical report – 70%

Please adhere to coding good practice

  - Meaningful variable names – please avoid names like “x”, “y”, or personal names
  - Using block comments to explain what your code does at the beginning and using line comments to clarify what this line of code does.
  - Consistent indentation – Indentation helps in making the code readable and easy to interpret. See an example of what to do and what to avoid, below:

Good practice ✓

```
#include “mbed.h”
char switch_word ;         //the word we will send
char recd_val;            //the received value
int main() {
     async_port.baud(9600);      //set baud rate to 9600
     recd_val = 0;
     while (1)
     {
        switch_word=0xa0;
     }
}
```

Avoid this ✗

```
#include “mbed.h”
char switch_word ;         //the word we will send
char recd_val;            //the received value
int main() {
```

```
async_port.baud(9600);          //set baud rate to 9600
recd_val = 0;
while (1)
{
switch_word=0xa0;
}
}
```

All the best of luck.

The Embedded Processors Team,

Brief:

The aim of the project is to design and develop a ***low power*** patient monitoring system capable of ***real-time*** temperature monitoring and fall detection, using the NUCLEO L432KC mbed microcontroller development board, the TMP102 digital temperature sensor, the ADXL345 accelerometer and other input/output peripherals that you may find necessary, e.g., push buttons, LEDs, buzzer, etc (***those will be your design choices***). By integrating sensors and alert mechanisms, the system ensures prompt response to abnormal temperature levels and fall events, enhancing patient safety and facilitating timely medical intervention.

The Assignment:

The aim of this assignment is to design, implement, and test an embedded system that operates as a ***Patient Temperature Monitoring and Fall Detection System***, using input/output peripherals connected to the NUCLEO L432KC board.

Detailed Description:

Patient CareGuard is to be designed to monitor patients' vital signs, i.e. temperature, and activity levels (for fall detection). The *TMP102* sensor ensures real-time temperature monitoring, enabling early detection of fever, while the *ADXL345* accelerometer detects sudden changes in acceleration indicative of falls. Upon detection of critical events, such as abnormal temperature levels or fall incidents, the system activates alert mechanisms such as audible alarms and/or visual indicators to promptly notify caregivers or medical personnel.

The **threshold** for the alarms to be generated and for falls is **for you to decide** on in your design and **a reasonable justification** for your design choices **must be clearly stated in your report**.

Hardware:

Required:

- Breadboard.
- One Nucleo L432KC mbed board and USB programming cable.

- One TMP102 sensor.
- One ADXL345 accelerometer.

One or more of the following (based on your design choice):

- Piezo buzzer.
- LEDs.
- LCD/OLED display.
- Push buttons.
- On/Off switch.
- Other components available in the lab that you may find necessary.

Software:

mbed online compiler (https://studio.keil.arm.com/auth/login/) or Desktop IDE – You will need to create an account to use the online compiler.

Task Splitting and Guidance:

- Take a modular approach to this design problem and breakdown your system into smaller components.

- Study the hardware and software requirements of each component and start to implement them, one at a time, e.g., start by interfacing the TMP102 and then the ADXL345, independently, or vice versa.

- Start to then work on the logic that will fulfill the system requirements, i.e., the temperature monitoring and fall detection system based on input from the TMP102 and ADXL345, respectively.

- Conduct unit testing of all your system components (hardware and software) to ensure that when you integrate them, you will have eliminated errors that can rise from them.

Please note that this is a design problem and not just a development exercise. You must:

- Carefully design your system based on the list of requirements. Justify your choices!
    - Take design decisions on the necessary components, threshold values for temperature alarm, threshold values for falls.
    - Clearly state any assumptions you made and justify them.
- Clearly outline the specifications of your system.
- Show the software and hardware design using flowchart(s) and system architecture/block diagrams.

Technical Report:

A formal written report is required that details the analysis, design, implementation, and testing carried out for this project.

Your report ***must*** cover and include the following:

1. A front page (***including the course title (UESTC2004 or UESTCHN2004), project title, your name and student GUID***)
2. A contents page showing the page numbers, list of tables, list of figures, etc.
3. **Introduction**: This section provides an overview of the project, including its purpose, objectives, and scope. It should also include a brief discussion of the background, with some literature references, and motivation for the project. **(Weight – 10 marks out of 70)**
4. **Methodology**: This section describes the methods used to design, implement, and test the embedded system. It should include a detailed description of the hardware and software used in the project, as well as any experimental procedures used to validate the system's functionality, including testing mechanisms. This is where you must highlight your design choices and justify them. **(Weight – 25 marks out of 70)**
5. **Results and Discussions**: This section presents the results of the project, including any data or observations gathered during testing. It should also include an interpretation of the results and how they relate to the project's objectives. **(Weight – 20 marks out of 70)**
6. **Conclusions and Future Work**: This section summarises the key findings of the project and discusses their implications and significance for the field of embedded systems engineering and healthcare. It should also highlight the project's contributions and potential impact. Lastly, it should discuss any limitations of the study and suggest areas for future work. **(Weight – 10 marks out of 70)**
7. **References**: This section lists all the sources cited in the report, using a consistent citation style (IEEE style). **(Weight – 5 marks out of 70)**
8. **Appendices**: This section includes any additional information that is relevant to the project but not included in the main text of the report, such as detailed technical specifications or codes.

Deliverables:

You are required to submit on Moodle *a single compressed file (.zip or .rar or a suitable format)*, that includes the following, by the ***31st of May 2024*:**

1. A formal written report as highlighted above.

2. A video demonstration of your system while in operation (Not too long - 3 to 5 minutes). (Please explain the operation with your voice during recording and make sure to do rigorous testing to showcase your system's full operation). – **(15 out of 30)**

3. All your C++ codes showing your detailed implementation with comments and compliance to good coding practice. – **(15 out of 30)**